
FORTIGATE NSE4

Lab 2 — Configuring FortiGate Interfaces, VLANs, Routing & Firewall Policies

A hands-on lab covering LAN interface configuration with DHCP services, VLAN segmentation concepts (router-on-a-stick vs. separate interfaces), static routing for internet access, firewall policy creation, NAT configuration, and end-to-end connectivity verification.

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Category: FortiGate NSE4 Lab Series · Portfolio Project

Platform: FortiGate VM v7.6.5 on EVE-NG · NAT/Routed Mode

1. Lab Topology

This lab extends the topology from Lab 1 by adding a second test PC (Test2) connected through a layer-2 switch. The switch enables VLAN segmentation scenarios between the two test clients.

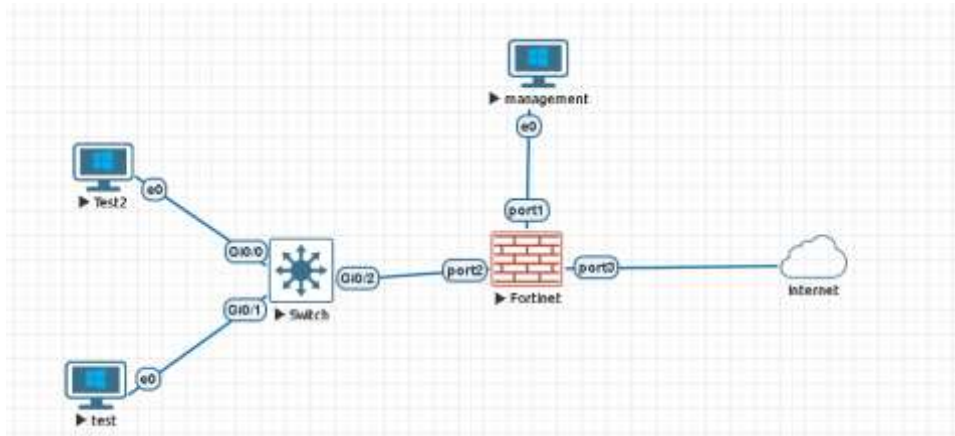


Figure 1 — Lab Topology: Management PC, two test PCs via switch, FortiGate, Internet

Port	Alias	Role	Connected To
port1	MGMT	Management	Management PC (192.168.100.x)
port2	LAN	Inside / LAN	Switch → Test PC + Test2
port3	WAN	Outside / Internet	Cloud (Internet)

2. LAN Interface Configuration

► 2.1 Configure port2 as LAN Gateway

Port2 acts as the default gateway for the LAN segment. Configure it with a static IP and enable a DHCP server so test clients receive addresses automatically:

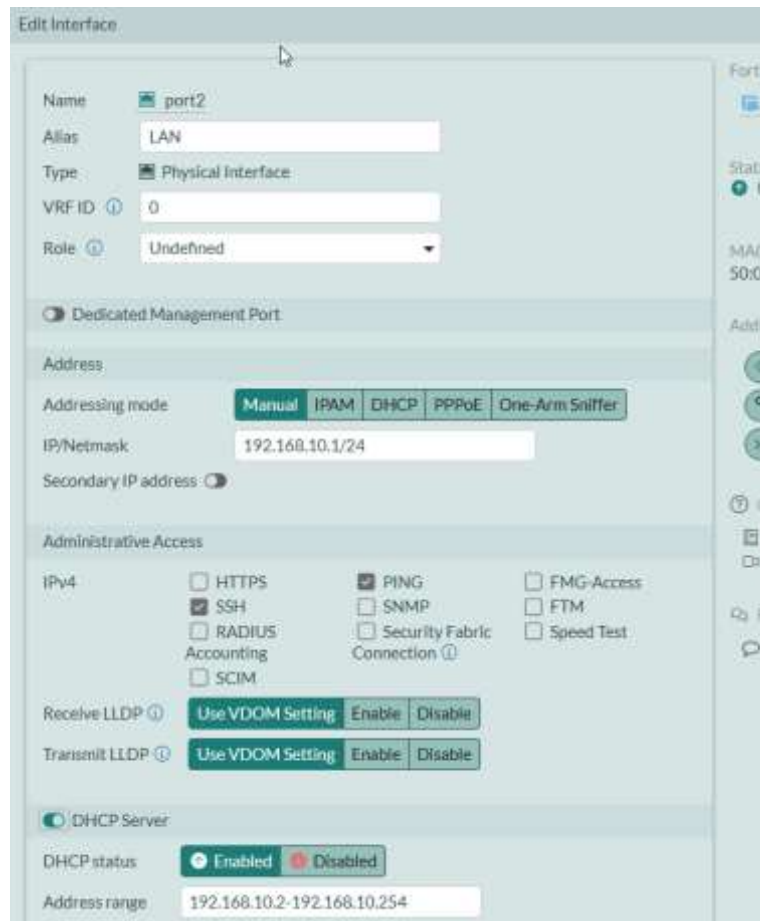


Figure 2 — Edit Interface: port2 (LAN) with DHCP server enabled

Setting	Value
Name	port2
Alias	LAN
Addressing mode	Manual (static)
IP/Netmask	192.168.10.1/24
Administrative Access	PING, SSH (no HTTPS — management is on port1)
DHCP Server	Enabled
DHCP Address Range	192.168.10.2 – 192.168.10.254

Note: Running DHCP on the firewall is acceptable for lab and small environments. In production OT networks, use a dedicated DHCP server for better control and auditing.

► 2.2 Verify DHCP Assignment

After configuring port2, both test machines (Test and Test2) receive IP addresses from the FortiGate DHCP pool with the correct gateway (192.168.10.1):

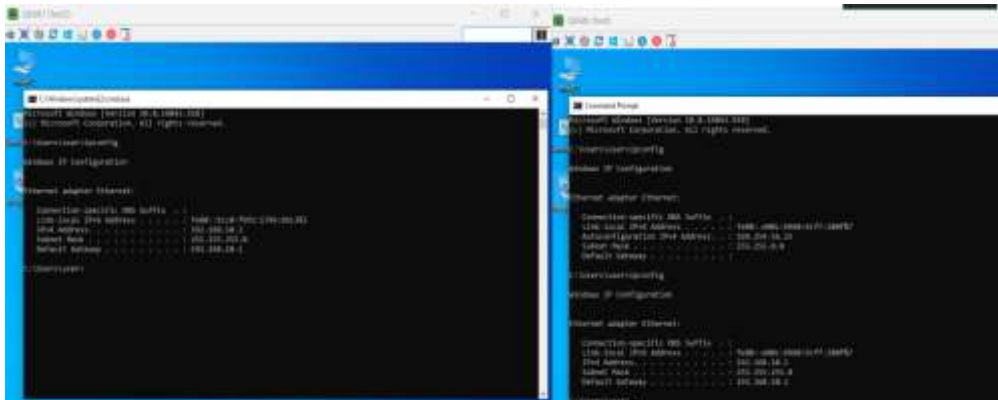


Figure 3 — Both test PCs received DHCP addresses (192.168.10.x)

3. VLAN Segmentation

► 3.1 The Problem — Isolating Hosts

With both test PCs on the same port2 interface, they share the same broadcast domain and DHCP pool. To isolate them into separate VLANs with different DHCP scopes, you need Layer 2 separation.

► 3.2 Key Concept: DHCP Separation = Layer 2 Separation

FortiGate binds DHCP scopes per interface. Each VLAN is a logical interface, so each VLAN gets its own DHCP pool. Without VLANs, you cannot have multiple DHCP scopes on the same physical port.

- DHCP is assigned per interface on FortiGate
- VLAN = logical interface (sub-interface)
- No VLAN = no multiple DHCP scopes

► 3.3 Preparing port2 as a Trunk

To use VLANs, first remove the IP address and disable DHCP on port2, converting it into a pure trunk interface:



Figure 4 — port2 as trunk: IP removed (0.0.0.0), PING/SSH access only

► 3.4 Evaluation Mode Limitation

In FortiGate evaluation mode, creating two VLANs on the same physical interface triggers a licensing error:

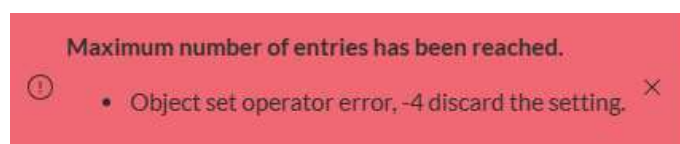


Figure 5 — "Maximum number of entries has been reached" error

Workaround: Use two separate physical interfaces (one per VLAN) and perform the segmentation at the switch level instead of router-on-a-stick.

► 3.5 Router-on-a-Stick (Licensed Mode)

In a fully licensed FortiGate, you can create VLAN sub-interfaces on the same physical port. Each VLAN sub-interface gets its own VLAN ID, IP address, and DHCP scope:

- Create new interfaces of type VLAN
- Assign VLAN IDs (e.g., 10 and 20)
- Bind each to the parent interface (port2)
- Configure a separate DHCP pool per VLAN

Switch Configuration (for Router-on-a-Stick)

Configure the switch port connected to FortiGate as a trunk allowing both VLANs, and set access ports for each test PC:

```
! Trunk to FortiGate
interface Gi0/2
    switchport mode trunk
    switchport trunk allowed vlan 10,20

! Access port for Test PC (VLAN 10)
interface Gi0/0
    switchport mode access
    switchport access vlan 10

! Access port for Test2 PC (VLAN 20)
interface Gi0/1
    switchport mode access
    switchport access vlan 20
```

Verification

On the FortiGate CLI, verify DHCP leases per VLAN:

```
# diagnose ip dhcp lease-list
```

Expected output: 192.168.10.x on VLAN10, 192.168.20.x on VLAN20.

4. Static Routing

Configure a default static route to direct all outbound traffic through the WAN interface (port3) toward the upstream gateway:

The screenshot shows the 'Edit Static Route' configuration window. The 'Destination' field is set to '0.0.0.0/0.0.0.0' with 'Subnet' and 'Internet Service' tabs. The 'Gateway Address' field is set to '192.168.1.1' with 'Dynamic' and 'Specify' tabs. The 'Interface' field is set to 'WAN (port3)'. The 'Administrative Distance' field is set to '10'. The 'Comments' field is empty. The 'Status' is set to 'Enabled'.

Figure 6 — Default static route: 0.0.0.0/0 via 192.168.1.1 (WAN/port3)

Field	Value
Destination	0.0.0.0/0.0.0.0 (default route)
Gateway Address	192.168.1.1 (upstream router/ISP)
Interface	WAN (port3)
Administrative Distance	10
Status	Enabled

5. Firewall Policy

FortiGate will NOT forward any traffic without an explicit firewall policy — this is a fundamental difference from traditional routers. Every traffic flow requires a matching policy.

► 5.1 Create LAN-to-Internet Policy

Navigate to Policy & Objects → Firewall Policy → Create New:



Figure 7 — Firewall policy: LAN (port2) → WAN (port3), ACCEPT, NAT enabled

Field	Value
Name	LAN_TEST_to_Internet
Incoming Interface	port2 (LAN)
Outgoing Interface	port3 (WAN)
Source	192.168.10.0/24 (or address object)
Destination	all
Service	ALL
Action	ACCEPT
NAT	Enabled (source NAT to outgoing interface IP)
Log Allowed Traffic	ALL

NAT must be enabled so that internal private addresses are translated to the WAN interface IP when reaching the internet. Without NAT, return traffic from the internet cannot reach the internal hosts.

► 5.2 Policy Processing Order

FortiGate evaluates policies top-down. If you have multiple policies, the order matters:

- More specific rules (block specific apps/URLs) should be placed above general allow rules
- A default implicit deny exists at the bottom — any traffic that does not match an explicit policy is dropped
- VLAN-specific policies (VLAN10 → WAN, VLAN20 → WAN) must each have their own policy

6. Verification

After deploying the static route and firewall policy, verify end-to-end internet connectivity from the test PC by pinging a public DNS server:

```
Pinging 8.8.8.8 with 32 bytes of data:  
Reply from 8.8.8.8: bytes=32 time=50ms TTL=119  
Reply from 8.8.8.8: bytes=32 time=48ms TTL=119  
Reply from 8.8.8.8: bytes=32 time=47ms TTL=119  
Reply from 8.8.8.8: bytes=32 time=47ms TTL=119
```

Figure 8 — Ping 8.8.8.8 successful: 0% loss, ~48ms average

Successful replies confirm the complete traffic path is operational: Test PC → port2 (LAN) → FortiGate routing → port3 (WAN) → NAT → Internet → return.

7. Summary

Component	Details
LAN Interface	port2 — 192.168.10.1/24 with DHCP (192.168.10.2–254)
WAN Interface	port3 — connected to Internet
DHCP Concept	One DHCP scope per interface; VLANs = separate logical interfaces
VLAN Approach	Router-on-a-stick (licensed) or separate interfaces (evaluation)
Static Route	0.0.0.0/0 → 192.168.1.1 via WAN (port3), AD 10
Firewall Policy	LAN → WAN, ACCEPT, NAT enabled, log all traffic
NAT	Source NAT (outgoing interface IP)
Implicit Deny	All unmatched traffic is dropped by default
Verification	Ping 8.8.8.8 from test PC — 0% loss
